

Techno-economic assessment and modeling of green hydrogen production using renewable energy: a case study of Moroccan regions

Omrane Derhy¹, Tasnime Bouanou^{1,*}, Nouhayla Atmani, Loubna Bousselamti, Meryeme Azaroual, Nouhaila Nabil, Ayoub Hirt

Research Institute on Solar Energy and New Energies (IRESEN), Morocco

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ABSTRACT

This paper presents a techno-economic and dynamic assessment of a renewable-powered green hydrogen production system intended for biofuel upgrading within the SUSTEPS project. The study designs an optimal system capable of continuously supplying 7.28 kg/h of hydrogen required for the biofuel production process, using photovoltaic and wind energy across four Moroccan regions. Unlike previous Moroccan hydrogen assessments, this work integrates green hydrogen production modeling directly into upgrading process system while combining regional optimization with dynamic operational validation. Techno-economic analysis using HOMER Pro identifies Dakhla–Oued Ed-Dahab as the most favorable region, achieving the lowest LCOH of 5.42 \$/kg at the studied scale. A scale-up analysis further demonstrates significant cost reduction potential. The optimized configuration is validated through MATLAB/Simulink dynamic modeling, confirming stable operation under variable renewable conditions and hydrogen production exceeding the nominal demand to ensure storage reserve. The results demonstrate the technical feasibility and economic relevance of integrating renewable hydrogen production into sustainable biofuel systems in Morocco.

1. Introduction

The rapid depletion of fossil fuel reserves, combined with the growing challenge of climate change, has driven the global energy sector toward a critical transition point, demanding a shift to cleaner and more sustainable energy systems [1]. Industrial activities, which account for nearly 19% of the world's total final energy consumption, still rely heavily on fossil energy carriers such as coal and natural gas [2]. This dependence contributes significantly to greenhouse gas emissions, highlighting the urgent need for decarbonized energy pathways to ensure environmental, economic, and social sustainability [3–7]. In recent years, renewable energy technologies, particularly solar photovoltaic (PV) and wind power, have played a pivotal role in facilitating this transition, providing flexible and economically viable alternatives to traditional energy systems. According to the International Energy Agency's Renewables 2025 report [8], Global renewable power capacity is projected to increase by 4600 GW, effectively doubling by 2030. Moreover, in over 80% of countries, the growth rate of renewable capacity between 2025 and 2030 is expected to exceed that of the previous

five years. In addition, the global energy transition is increasingly driven by rapid growth in renewable electricity deployment, with solar PV and wind technologies leading capacity additions worldwide.

Fig. 1 illustrates annual renewable electricity capacity additions by technology from 2019 to 2025.

It highlights the dominant role of solar PV, both utility scale and distributed, which accounts for the largest share of new capacity, followed by wind power. This trend highlights the rapid acceleration of clean electricity generation and supports the feasibility of using solar PV and wind to power green hydrogen systems as a major decarbonization route.

In the same context, several studies all over the world have investigated hydrogen production using renewable energy sources and evaluated the associated system costs [9,10]. In [11], the authors assessed solar and wind based green hydrogen production across Africa through a techno-economic and environmental analysis. The results showed that solar resources provide higher hydrogen yields than wind, with Namibia reaching 8.54 kg/m² and achieving one of the lowest LCOH values (4.60 \$/kg), closely followed by Egypt (4.64 \$/kg). PV systems exhibited

* Corresponding author.

E-mail addresses: bouanou@iresen.org, bouanou.t@gmail.com (T. Bouanou).

¹ These authors contributed equally to this work.

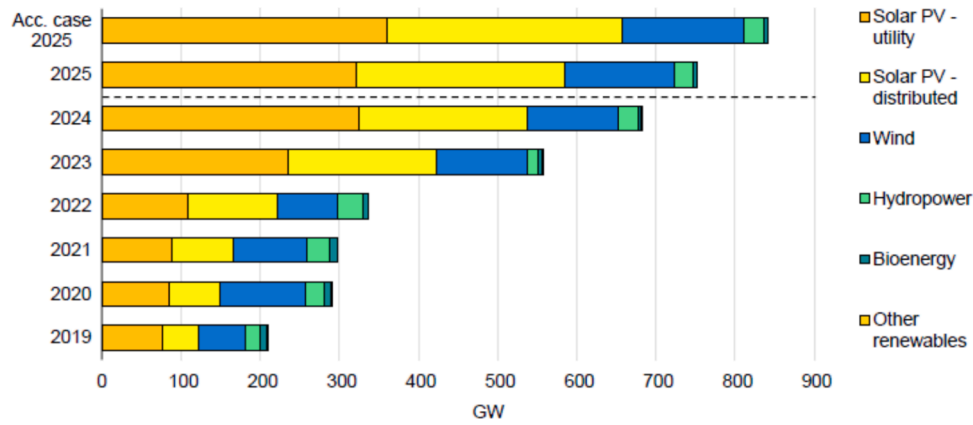


Fig. 1. Renewable electricity capacity additions by technology, 2019–2025 [8].

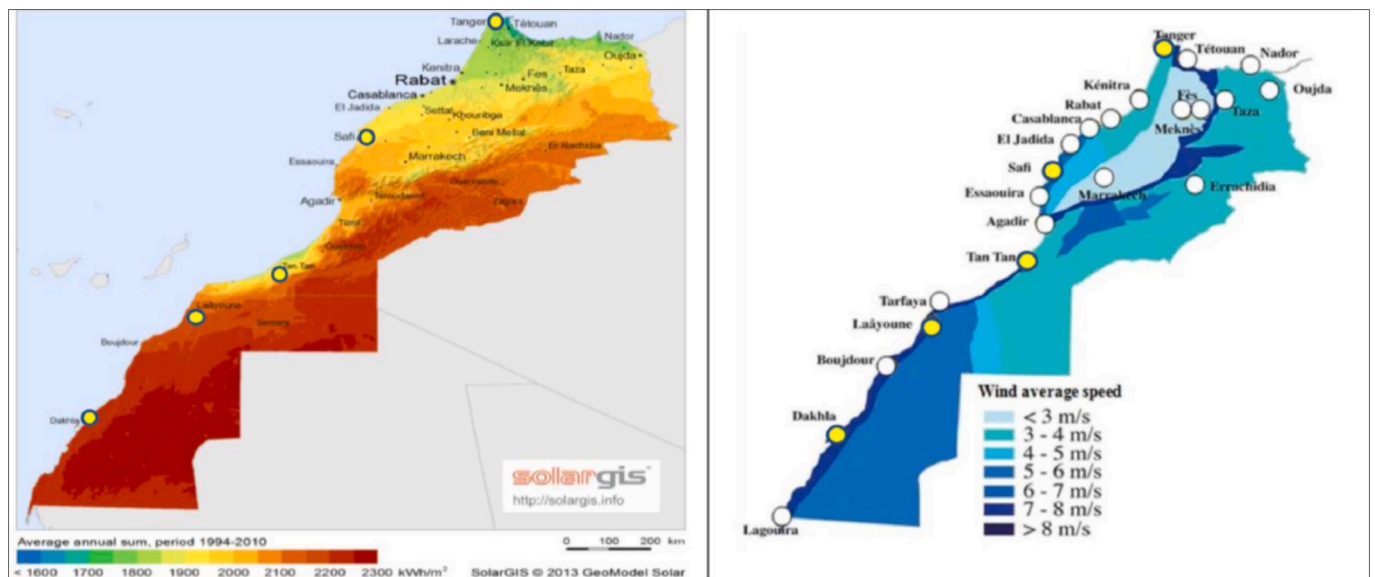


Fig. 2. PV and Wind potential in Morocco [22].

overall lower hydrogen costs (4.6–7.31 \$/kg), whereas wind systems were generally more expensive, often exceeding 20 \$/kg. Simulations using Matlab Simulink further indicated that PV powered electrolysis offers higher system efficiency, while wind based production suffers from stronger intermittency. [12] evaluated the wind energy potential for electricity generation and green hydrogen production in Djibouti across five candidate sites. Their analysis showed competitive hydrogen costs, with the LCOH ranging from 1.79 to 3.38 \$/kg, while the planned 450 MW wind capacity could generate up to 1739 GWh/year. Also [13] assessed the solar potential of Morocco for green hydrogen production using both low and high temperature electrolysis. The results showed that regions such as Dakhla and Laâyoune offer the highest hydrogen output due to superior solar resources. The study reported that low temperature electrolysis yields hydrogen production costs between 4.5 and 8.7 \$/kg, while high temperature systems can reduce the cost to around 3.4 \$/kg. [14] developed an analytical techno-economic model to assess green hydrogen production from a PV power station in Salalah, Oman. The study showed that a 5 MWp PV plant can produce around 90,910 kg of hydrogen per year, with an LCOH of about 6.2 \$/kg. In [15], authors conducted a multi criteria analysis to select the most suitable clean technology for industrial scale hydrogen production in Morocco. Ten production routes were evaluated using the AHP and Fuzzy VIKOR methods, and the results revealed that alkaline water

electrolysis powered by renewable energy sources is the most appropriate option for the Moroccan context, with an estimated hydrogen production potential of about 1057.26 Mt/year using solar PV. [16] evaluated green hydrogen production using a hybrid PV/wind system under different climatic conditions in Egypt. The annual hydrogen production ranged from 1,196 to 1,972 kg, with an overall system efficiency of 7.69–9.37% and a levelized cost between 4.54 and 7.48 \$/kg. [17] compared hydrogen production from solar energy in Benguerir (Morocco) and Almeria (Spain) using a 100 MWp PV plant coupled with a PEM electrolyzer. The results showed higher annual hydrogen production in Morocco (3342 tons) than in Spain (3230 tons), with a lower LCOH of 5.78 \$/kg, confirming the stronger potential of Morocco for cost-effective green hydrogen production. [18] evaluated a green methanol plant in Ordos, China, and optimized renewable hydrogen production using solar and wind energy. Their results showed that a hybrid solar/wind configuration is the most cost-effective option, and that wind turbine costs strongly affect overall economics.

Although several studies have assessed the potential of green hydrogen production in Morocco, most focus mainly on resource assessment or separate techno economic evaluation of renewable powered electrolysis systems. In contrast, the present work combines detailed MATLAB/Simulink dynamic modeling with HOMER Pro techno economic optimization, includes hydrogen storage behavior, and

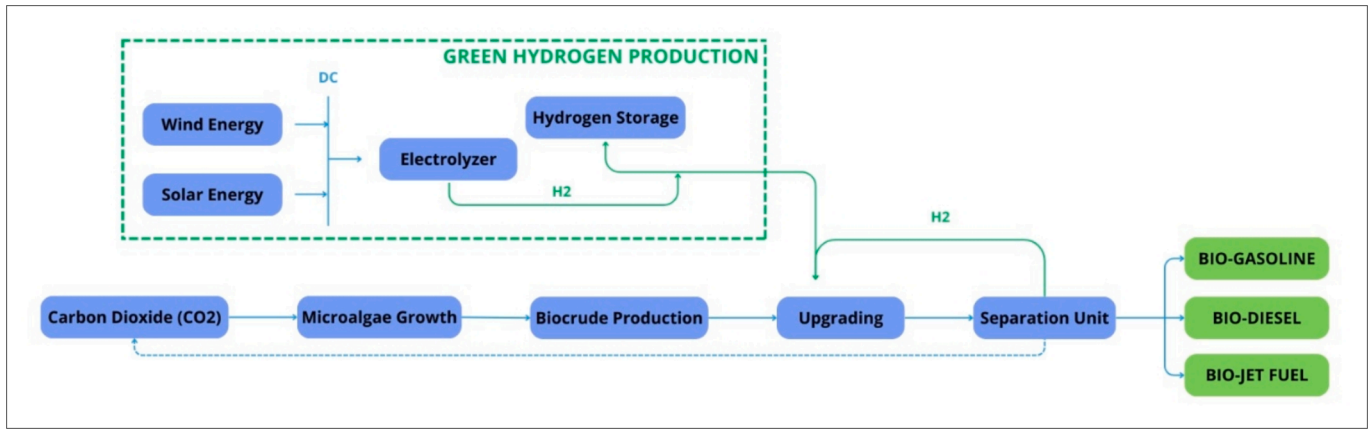


Fig. 3. Simplified Scheme of the SUSTEPS Project: Focus on Green Hydrogen Production. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

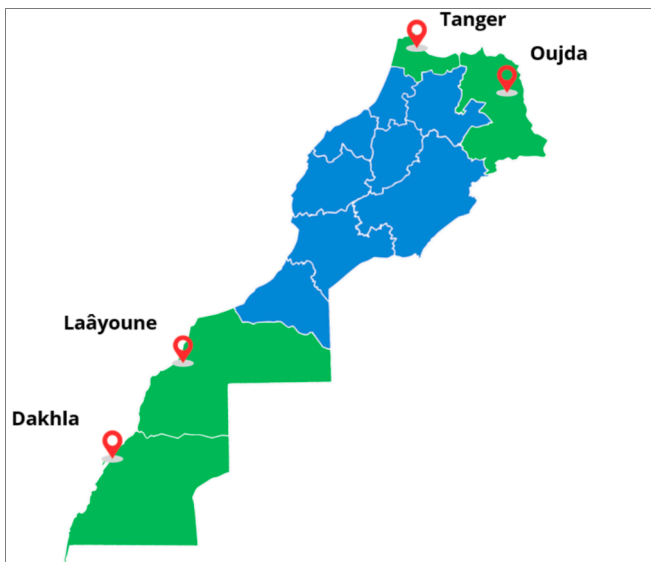


Fig. 4. The four selected regions of Morocco (green color). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 1

Meteorological attributes of the four study sites [32].

	Dakhla	Laâyoune	Oujda	Tanger
Latitude (°N)	23°59.3	27°1.1	34°39.7	35°45.1
Longitude (°W)	15°39.7	13°24.9	1°59.2	5°42.3
Annual average Temperature (°C)	20.6	20.31	16.56	17.88

examines system scalability across different Moroccan regions within the SUSTEPS project. This combined modeling and scaling approach represents the main contribution of this study.

Morocco has very strong solar and wind resources that complement each other geographically. The best solar potential is in the southern and southeastern regions, while the strongest wind resources are along the Atlantic coast (north and south) and in some inland passes [19]. These favorable conditions make the country well suited for the deployment of renewable energy technologies.

Regions with high solar PV potential include Drâa-Tafilalet, Souss-Massa, Laâyoune-Sakia El Hamra, and Dakhla-Oued Eddahab, where irradiation levels are among the highest in the country [20]. In contrast,

regions with strong wind potential include Tanger-Tetouan-Al Hoceima, Guelmim-Oued Noun, Laâyoune-Sakia El Hamra, and Dakhla-Oued Eddahab, especially along the Atlantic coast and specific inland corridors [21]. Fig. 2 illustrates the solar and wind energy potential in Morocco.

Morocco has made significant investments in both solar and wind energy as part of a long-term strategy to increase its share of renewables, reduce dependence on fossil fuels, and meet ambitious climate goals. Morocco hosts a number of large-scale solar and wind projects that highlight its commitment to renewable energy. In the solar sector, key developments include the Noor Ouarzazate CSP complex (580 MW, using parabolic trough and solar tower technology) [23], the Noor Midelt II and III projects (400 MW each, solar PV with battery storage) [24] and the Aïn Beni Mathar ISCC plant (472 MW, combining CSP and natural gas) [25]. On the wind side, major projects include the Tarfaya Wind Farm (301 MW) [26], the Jbel Lahdid Wind Farm (270 MW) near Essaouira [27], the Aftissat Wind Farm (201.6 MW, with planned expansion to 401.6 MW) near Boujdour [28], and the Boujdour Wind Farm (300 MW) [29]. These projects collectively demonstrate Morocco's rapid expansion of renewable capacity and its strategic use of solar and wind resources across different regions [30], highlighting the importance of collaboration among government, industry, and academia. In addition, large scale deployment of renewable hydrogen systems relies on supportive regulations, including energy integration, safety standards, and grid access as well as regional socio economic conditions and local implementation contexts. Clear and stable policies are essential to attract investment and speed commercialization.

The present study is conducted within the framework of the international SUSTEPS project, which aims to advance sustainable microalgae-based biofuel production. The project covers the full value chain of this emerging technology including microalgae cultivation and CO₂ fixation, conversion of biomass into biocrude, biocrude upgrading, green hydrogen production, and the valorization of aqueous streams [31]. Fig. 3 illustrates the simplified scheme of the project. The section outlined within the frame in the figure is the focus of our design in this paper.

Biofuels are a key part of the energy transition, providing a renewable alternative for sectors that are hard to electrify, such as aviation, heavy transport, and industry. Hydrogen plays a crucial role in upgrading processes, where it is used in hydrocracking to transform complex and unstable biocrude into lighter, stable fuels compatible with existing petroleum infrastructure, acting as a reducing agent with a catalyst to guide the reactions. This enables the production of fuels such as bio-diesel, bio-jet fuel, and bio-gasoline...

The present paper focuses on the green hydrogen production section of the project by modeling a complete green hydrogen production and

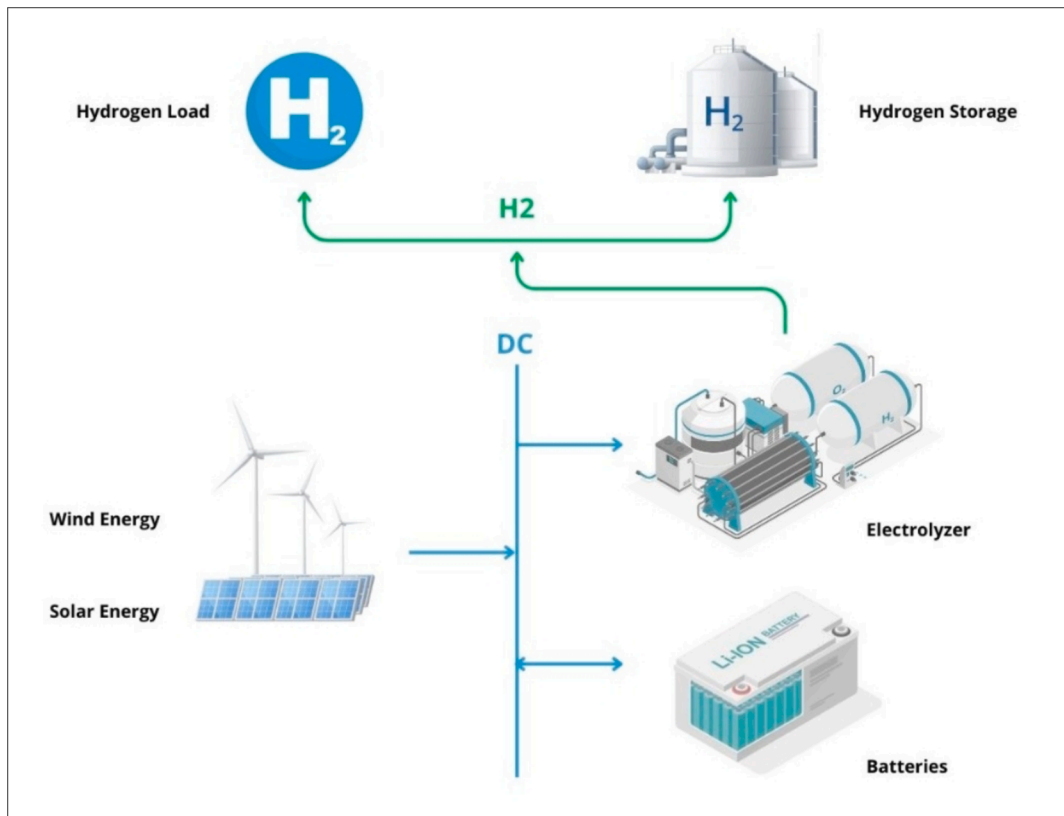


Fig. 5. Structural Overview of a Solar-Wind Green Hydrogen Production System. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

storage chain powered by renewable energy sources.

The hydrogen production system is designed to operate in direct integration with the biofuel upgrading process within the same industrial zone considered in the SUSTEPS project. As a result, hydrogen is assumed to be consumed on-site, and no external transportation (pipelines or trucking) is required.

The study covers four regions of Morocco: Tanger-Tétouan-Al Hoceïma, L'Oriental, Laâyoune-Sakia El Hamra, and Dakhla-Oued Ed-Dahab. Fig. 4 illustrates the four selected regions of Morocco, and Table 1 provides an overview of the geographic attributes of the four study sites.

The project requires a total of 11.44 kg/h of hydrogen for the upgrading process. Of this, 4.16 kg/h is supplied by hydrogen generated internally from reforming the biogas produced during the hydrothermal liquefaction of microalgae (see separation unit in the figure). The remaining 7.28 kg/h must come from electrolysis powered by renewable energy sources, designed to operate continuously 24 h per day. The goal of this project is therefore to design an optimal system capable of continuously producing 7.28 kg/h of renewable hydrogen while considering technical, environmental, and economic aspects. Specifically, the study aims to:

- Determine the optimal energy mix for hydrogen production in four selected Moroccan regions.
- Estimate the associated costs, including LCOH and NPC in order to identify the most suitable region in terms of efficiency and economic viability to support the transition of Morocco toward a clean and sustainable energy alternative.
- Validate the proposed system by performing dynamic simulations to confirm its feasibility before the implementation phase.

This paper is structured into several sections. Section 2 presents the

methodology, including the system layout and description, the techno-economic analysis using HOMER Pro, and system modeling via MATLAB/Simulink. Section 3 presents the results and discussion of the HOMER Pro and MATLAB/Simulink simulations. Section 4 provides the conclusions.

2. Methodology

2.1. System layout and description

This section provides an overview of the system architecture. As shown in Fig. 5, green hydrogen is generated using energy from both solar panels and wind turbines. The system is composed of a renewable power source, an Alkaline Electrolyzer, and a hydrogen storage unit.

An electrolyzer is an electrochemical device that converts electrical energy into chemical energy by driving the water electrolysis reaction, in which water is split into hydrogen and oxygen gases using an external electric current.

Based on their operating principles and materials, electrolyzers can be divided into four main categories: alkaline, PEM, AEM, and SOEC. Among these technologies, the alkaline electrolyser is adopted in this project. The selection of alkaline electrolysis technology is motivated by its high technological maturity, lower capital cost per kW compared to PEM and SOEC systems, and proven reliability in large scale industrial hydrogen production, making it particularly suitable for applications requiring continuous hydrogen supply, as in the SUSTEPS upgrading process.

The required electrolyzer capacity to produce a desired hydrogen mass flow rate can be estimated using the energy balance of the electrolysis process. The mass flow rate of hydrogen (m_{H_2}) is related to the installed electrolyzer capacity (I_0) by [33]:

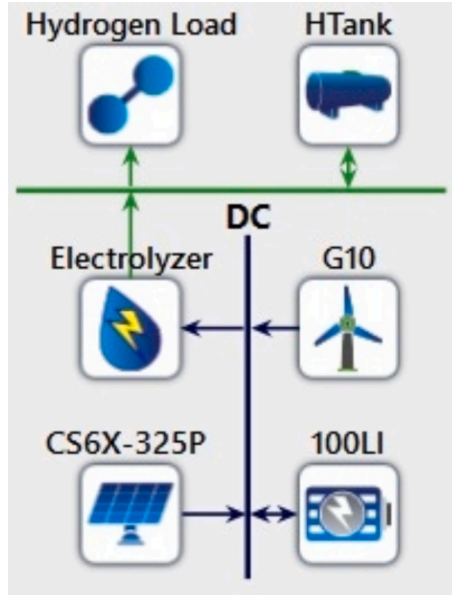


Fig. 6. System Architecture Designed Using HOMER Pro.

Table 2
Wind Speed and Solar Irradiation Data for the Four Selected Cities [32].

Regions	Wind Speed (m/s)	Irradiation (Kwh/m ² /day)
Tanger–Tétouan–Al Hoceima (Tanger)	5.59 m/s	5.04 Kwh/m ² /day
Oriental (Oujda)	4.90 m/s	4.92 Kwh/m ² /day
Laâyoune–Sakia El Hamra (Laâyoune)	7.47 m/s	5.97 Kwh/m ² /day
Dakhla–Oued Ed-Dahab (Dakhla)	7.86 m/s	5.76 Kwh/m ² /day

Table 3
Financial parameters of the system elements 22,34.

Component	Operations & maintenance cost per year	Capital	Replacement
PV panel (325 W/unit)	14 (\$/kW)	750 (\$/kW)	700 (\$/kW)
Wind turbine (10KW/unit)	500 (\$/unit)	17 500 (\$/unit)	17 500 (\$/unit)
Electrolyzer	120 (\$/kW)	1200 (\$/kW)	1000 (\$/kW)
Hydrogen Tank	16 (\$/kg)	800 (\$/kg)	800 (\$/kg)
Battery (100 KWh Li-Ion)	500 (\$/unit)	20 000 (\$/unit)	20 000 (\$/unit)

$$m_{H_2} = \frac{I_C \times CF \times \varphi_{EL}}{HHV_{H_2}} \quad (1)$$

- where φ_{EL} is the system efficiency of the electrolyzer,
- HHV_{H_2} is the higher heating value of hydrogen in kWh/kg,
- CF is the capacity factor of the electrolyzer,
- is I_C the installed capacity in kW.

In addition, the energy required to power the electrolyzer is supplied by renewable energy sources, namely PV panels and wind turbines. Photovoltaic panels generate electricity by converting solar radiation into electrical energy through the photovoltaic effect. Incident photons create an electric potential in semiconductor materials, resulting in the production of electrical power in the form of direct current (DC). This DC power can then be conditioned as required to meet the electrolyzer's

electrical input specifications.

Moreover, wind turbines generate electricity by capturing the kinetic energy of the wind flow and converting it into mechanical energy through the rotation of the turbine blades. This mechanical energy drives a generator, which produces electrical power. The amount of electricity generated depends mainly on wind speed and turbine characteristics. In this study, the wind energy system is assumed to be equipped with a rectifier that converts the generated electrical output into direct current to supply the electrolyzer.

A hydrogen storage unit is an essential component of the overall energy system. The storage unit allows excess hydrogen produced by the electrolyzer during periods of high renewable generation to be safely stored for later use. This capability is important for balancing supply and demand, ensuring that the system can continuously provide energy even when renewable output fluctuates.

2.2. Techno-economic analysis of the system using HOMER Pro

The studied system is modeled using HOMER Pro software (order number#50093). This software is a simulation and optimization tool for the design and analysis of hybrid energy systems. HOMER Pro provides the capability to model multiple energy resources, including renewable and conventional sources, energy storage systems, and hydrogen production units. It performs simulations over the project lifetime, evaluates system performance under various operational scenarios, and identifies optimal configurations based on economic and technical criteria such as the LCOH and NPC. In addition, it determines the optimal renewable energy mix, detailed component sizing, energy flows, and overall system operation. The main objective of this study is to identify the region that offers the lowest LCOH and NPC while ensuring technical feasibility.

For the simulation, the system is designed to supply a continuous hydrogen flow rate of 7.28 kg/h to meet the requirements of the upgrading process within the global SUSTEPS project, operating continuously 24 h per day.

It is assumed that the wind turbines are equipped with integrated AC/DC converters. The analysis focuses exclusively on the main components directly involved in green hydrogen production, and thus the electrical consumption of auxiliary or secondary equipment is not taken into account. Fig. 6 illustrates the design architecture of a hydrogen system powered by mixed renewable sources, as generated by HOMER Pro.

The temporal variability of solar irradiance and wind speed was incorporated using hourly meteorological data for each considered region within the HOMER Pro environment. The simulation was conducted over an annual time horizon (8760 h), allowing seasonal and daily fluctuations in renewable resource availability to be captured explicitly. HOMER Pro performs time series simulation to evaluate system performance under variable generation conditions, ensuring that renewable intermittency and its impact on electrolyzer operation are reflected in the optimization results. Excess renewable energy during high production periods is either directed to hydrogen production or stored in the hydrogen tank, while periods of low renewable availability are compensated by previously stored hydrogen. This approach allows the model to account for short term and seasonal variability without assuming constant resource conditions.

To provide an overview of the climatic conditions in the four regions, the cities of Tangier, Oujda, Laâyoune, and Dakhla were selected to represent the regions of Tanger–Tétouan–Al Hoceima, Oriental, Laâyoune–Sakia El Hamra, and Dakhla–Oued Ed-Dahab, respectively. Table 2 presents the wind speed and solar irradiation data for these cities based on NASA data [32].

The table shows that Dakhla has the highest wind speed and the highest solar irradiation, making it the most favorable location for both wind and solar energy potential. Overall, the southern regions (Dakhla and Laâyoune) demonstrate superior renewable energy potential compared to the northern and eastern regions.

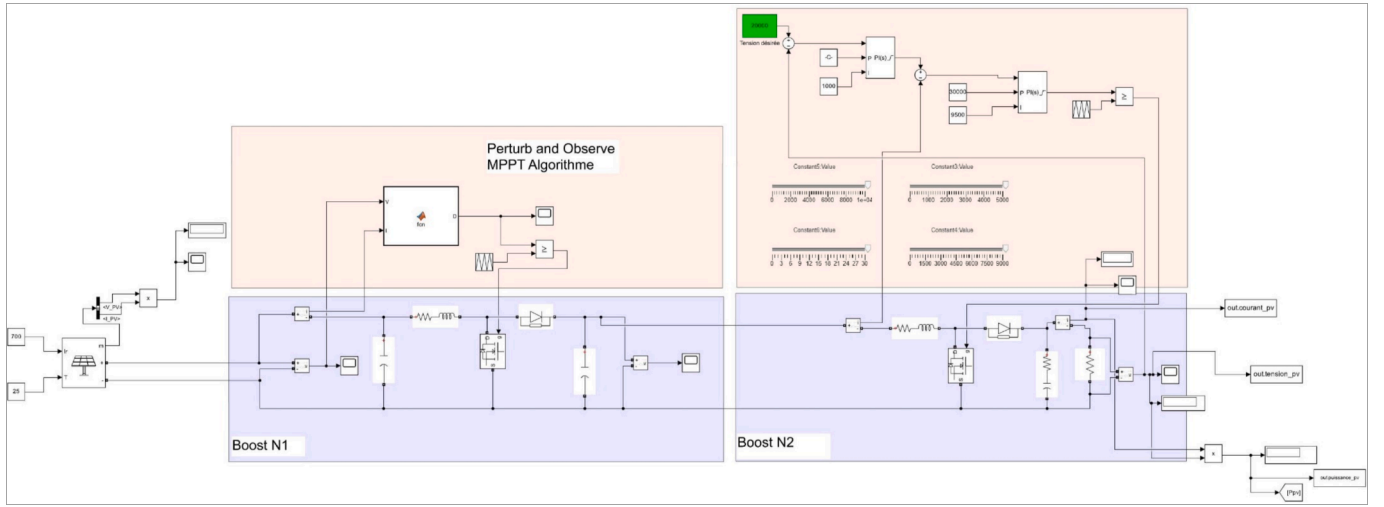


Fig. 7. Photovoltaic System Chain in MATLAB/Simulink.

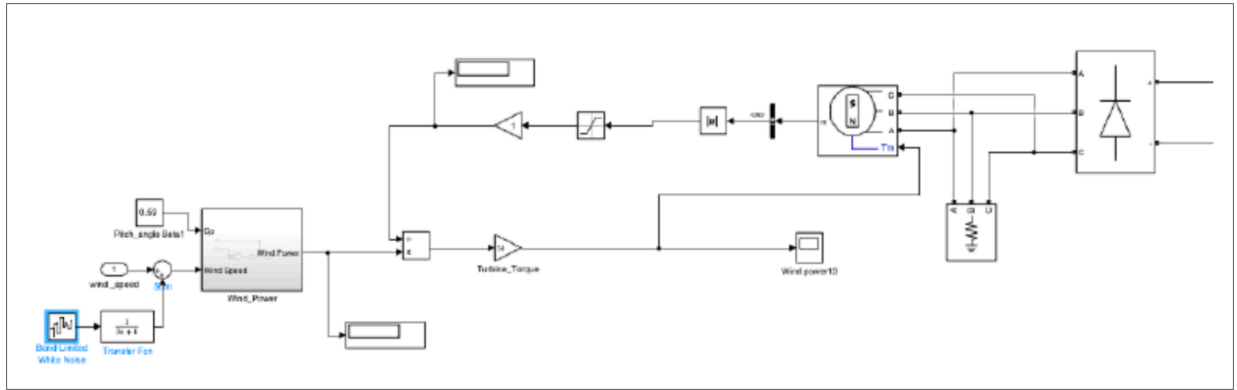


Fig. 8. Modeling of the Wind Turbine in MATLAB/Simulink.

2.2.1. Detailed economic cost breakdown

The economic assessment of the proposed hydrogen production system considers capital investment, replacement costs, and operation and maintenance expenses over the project lifetime. Table 3 summarizes the financial parameters used for each system component, including photovoltaic panels, wind turbines, electrolyzer units, hydrogen storage tanks, and battery systems. All cost values presented in Table 3 are expressed in USD and correspond to 2025 price levels to ensure currency and temporal consistency.

The initial capital investment is primarily driven by the electrolyzer and renewable generation units, which represent the largest share of total system cost. Operation and maintenance costs are relatively moderate compared to capital investment, indicating that the economic performance of the system is largely influenced by upfront installation costs rather than operational expenses. In addition, replacement costs are included to account for component lifetime limitations within the project duration, particularly for the electrolyzer and battery systems. However, the optimization results indicate that battery storage is excluded in the optimal configuration due to its high net present cost compared to hydrogen storage.

The economic assessment of the project is based on a total project lifetime of 25 years. Component lifetimes and degradation assumptions are defined as follows: the electrolyser and battery storage systems are assumed to have operational lifetimes of 15 years, while the hydrogen tank, wind turbine, and PV system are each assumed to operate over the full 25-year project horizon. A nominal discount rate of 8% is applied, corresponding to a real discount rate of 5.88%, based on an expected

inflation rate of 2%. Regarding hydrogen compression and storage, the CAPEX is assumed to be 800 USD/kg, with an equivalent replacement cost of 800 USD/kg and operation and maintenance costs estimated at 16 USD/kg, as detailed in Table 3. Moreover, within the HOMER Pro simulation framework, load management is handled through hourly energy balance calculations over the annual time horizon. When renewable generation exceeds the power required for electrolysis, surplus energy is directed to hydrogen production and stored in the hydrogen tank. Conversely, during periods of reduced renewable availability, previously stored hydrogen compensates for production deficits to maintain supply continuity.

2.3. System modeling via MATLAB Simulink

The purpose of the MATLAB Simulink model is to provide dynamic validation of the optimized system obtained from HOMER Pro simulations. While HOMER Pro performs techno-economic optimization based on hourly energy balances, it does not capture transient behavior or short-term system dynamics. Therefore, MATLAB Simulink is used to analyze the real time operation of the system under variable renewable inputs.

2.3.1. Modeling of the photovoltaic system

Fig. 7 presents the MATLAB/Simulink (version: 9.13.0 (R2022b)) model of the photovoltaic (PV) system developed for the Dakhla–Oued Ed-Dahab region. The model represents the electrical behavior of the PV array under region-specific solar irradiance and ambient temperature

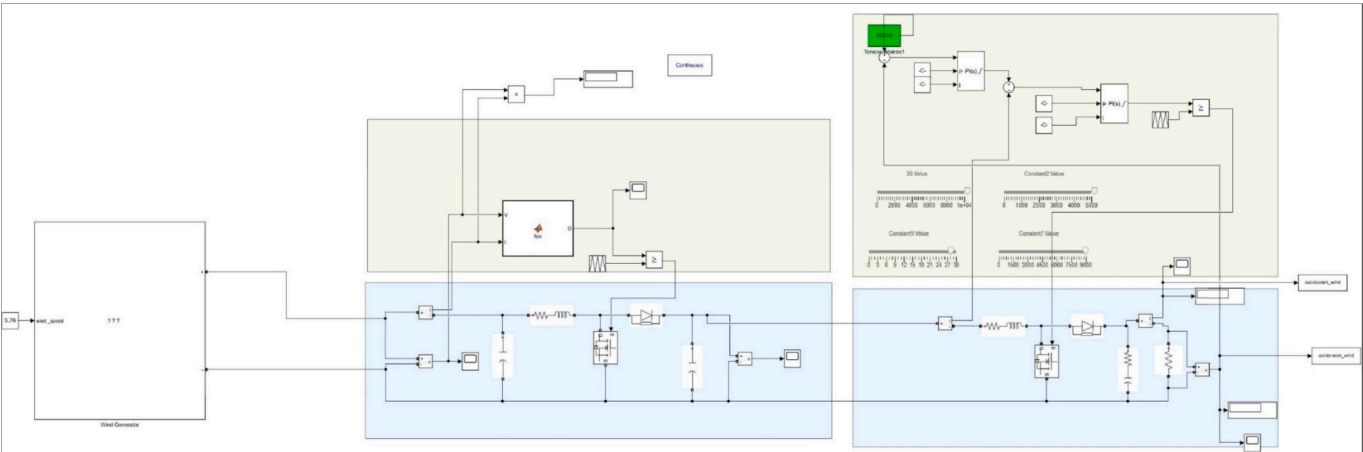


Fig. 9. Modeling of the Wind Turbine in MATLAB/Simulink.

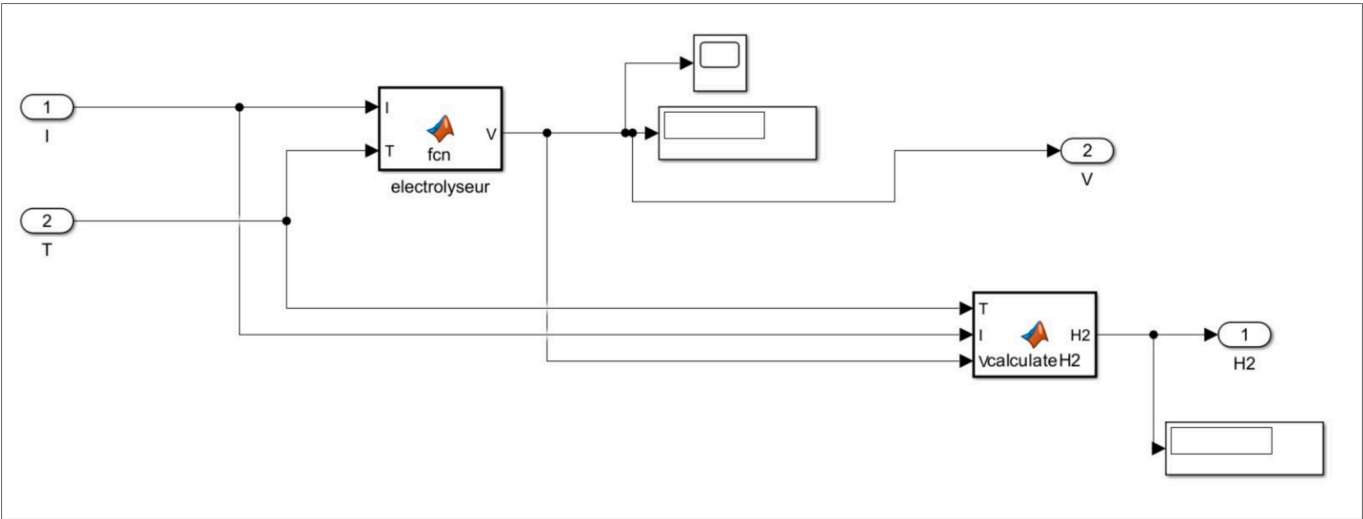


Fig. 10. Modeling of the alkaline electrolyzer in MATLAB/Simulink.

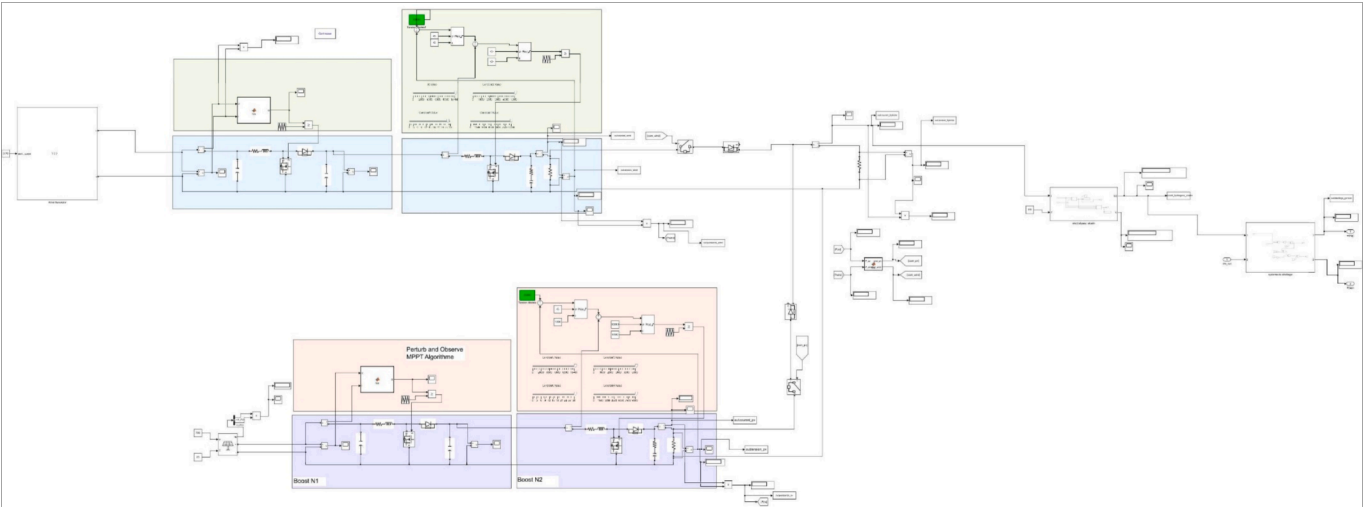


Fig. 11. Modeling of the complete hydrogen production chain.

conditions. It consists of two cascaded DC–DC boost converters, the first of which is combined with a Perturb and Observe (P&O) Maximum Power Point Tracking (MPPT) algorithm to extract the maximum available power from the PV generator. The P&O method operates by

Table 4
Optimal Results proposed by HOMER pro regarding the four regions.

City	PV (MW)	Wind turbines capacity (MW)	Number of wind turbines	Electricity production (Gwh/yr)
Dakhla–Oued Ed-Dahab (Dakhla)	0.785	0.6	60	3.52
Laâyoune–Sakia El Hamra (Laâyoune)	1.04	0.44	44	3.43
Oriental (Oujda)	2.526	0.06	6	4.31
Tanger–Tétouan–Al Hoceima (Tanger)	1.8	0.27	27	3.56

Table 5
Energy mix of the studied regions.

Location	PV (%)	Wind (%)
Dakhla–Oued Ed-Dahab (Dakhla)	56.7	43.3
Laâyoune–Sakia El Hamra (Laâyoune)	70.3	29.7
Tanger–Tétouan–Al Hoceima (Tanger)	87	13
Oriental (Oujda)	97.7	2.3

Table 6
Results of LCOH for the four regions.

Regions	LCOH (\$/Kg)
Tanger–Tétouan–Al Hoceima (Tanger)	6.86 (\$/Kg)
Oriental (Oujda)	7.24 (\$/Kg)
Laâyoune–Sakia El Hamra (Laâyoune)	5.57 (\$/Kg)
Dakhla–Oued Ed-Dahab (Dakhla)	5.42 (\$/Kg)

Table 7
Results of NPC for the four regions.

Regions	NPC (million \$)
Tanger–Tétouan–Al Hoceima (Tanger)	5.02 (million \$)
Oriental (Oujda)	5.37 (million \$)
Laâyoune–Sakia El Hamra (Laâyoune)	4.13 (million \$)
Dakhla–Oued Ed-Dahab (Dakhla)	4.02 (million \$)

introducing a small perturbation in the PV operating voltage and observing the resulting variation in output power. If the power increases following the perturbation, the control algorithm continues in the same direction; otherwise, the perturbation direction is reversed. This iterative process allows the operating point to converge toward the maximum power point under varying irradiance conditions.

The output of this stage feeds a second DC/DC boost converter regulated by PI controller, whose role is to stabilize and amplify the output voltage. The PI controller continuously adjusts the duty cycle of the converter based on the voltage error between the reference DC bus voltage and the measured output voltage. The proportional term ensures fast dynamic response, while the integral term eliminates steady-state error, resulting in stable voltage regulation suitable for electrolyzer operation.

2.3.2. Modeling of the wind energy system

The modeling process starts with the wind turbine representation using a wind profile defined by the average wind speed combined with a random signal to represent wind speed variations, as shown in Fig. 8. Following the wind turbine modeling, the complete system architecture was developed.

The wind energy system chain has been modeled in MATLAB/Simulink to simulate its dynamic behavior and integration within the hydrogen production chain. The Fig. 9 illustrates the complete configuration of the wind energy conversion system, including the wind

turbine, power electronic interfaces, and control blocks. It consists of the modeled wind turbine coupled with an electrical generator, an AC/DC rectifier responsible for converting the generated alternating current into direct current, and a first DC/DC converter dedicated to maximum power extraction through a maximum power point tracking (MPPT) algorithm. The controller adjusts the electrical load via this converter to match the turbine's optimal operating point, thereby maximizing energy capture under fluctuating wind speeds. Finally, a second DC/DC boost converter is added to regulate and stabilize the output DC voltage.

2.3.3. Modeling of the alkaline electrolyzer

Fig. 10 illustrates the MATLAB/Simulink model developed to represent the dynamic behavior of the alkaline electrolyzer used for hydrogen production. The electrolyzer block receives the electrical input power supplied by the renewable energy sources, along with the operating temperature, as input variables. Based on these inputs, the model computes the operating voltage of the electrolyzer and the corresponding hydrogen production rate. This model allows the assessment of electrolyzer performance under variable renewable energy supply conditions.

2.3.4. Modeling of the complete hydrogen production system chain

Following the modeling of each component, the entire system was integrated to form the complete hydrogen production chain as illustrated in Fig. 11.

2.4. Environmental impact considerations

Green hydrogen produced via electrolysis powered by photovoltaic and wind energy systems offers significant environmental advantages compared to conventional hydrogen production from natural gas reforming. By eliminating fossil fuel combustion during the hydrogen production phase, the system substantially reduces direct CO₂ emissions and contributes to decarbonization objectives. In addition to carbon emission reduction, the proposed system minimizes air pollutants and supports sustainable energy transition strategies. Overall, the integration of solar and wind energy in hydrogen production represents a low-carbon and environmentally favorable pathway aligned with Morocco's renewable energy strategy and global decarbonization goals.

However, the system demands significant capital investment because of the required renewable generation units, power electronics, electrolyzer infrastructure, and hydrogen storage systems.

3. Results and discussions

3.1. Results and discussions from HOMER PRO

The simulation results obtained from HOMER Pro for the four regions are summarized in Table 4, showing that renewable capacities, and system configurations vary depending on the site location, cost and characteristics. The results indicate that, in order to achieve a hydrogen production of 7.28 kg/h, the optimization software proposes different PV and wind turbines capacities for each region based on their specific climatic conditions and system cost minimization.

3.1.1. Renewable energy mix by regions

The optimal solution from HOMER Pro suggests a PV and wind energy mix, with the percentage of each differing across regions. Table 5 shows the energy mix of the studied regions. Based on the results, Dakhla has the most balanced distribution between PV and wind energy, with 56.7% PV and 43.3% wind, reflecting a significant contribution from both sources. Laâyoune relies more heavily on PV, which accounts for 70.3% of the mix, while wind contributes 29.7%. In contrast, northern regions such as Oriental (Oujda) and Tanger–Tétouan–Al Hoceima (Tanger) show a clear dominance of photovoltaic energy in the optimal configuration. Although areas such as Tanger possess strong

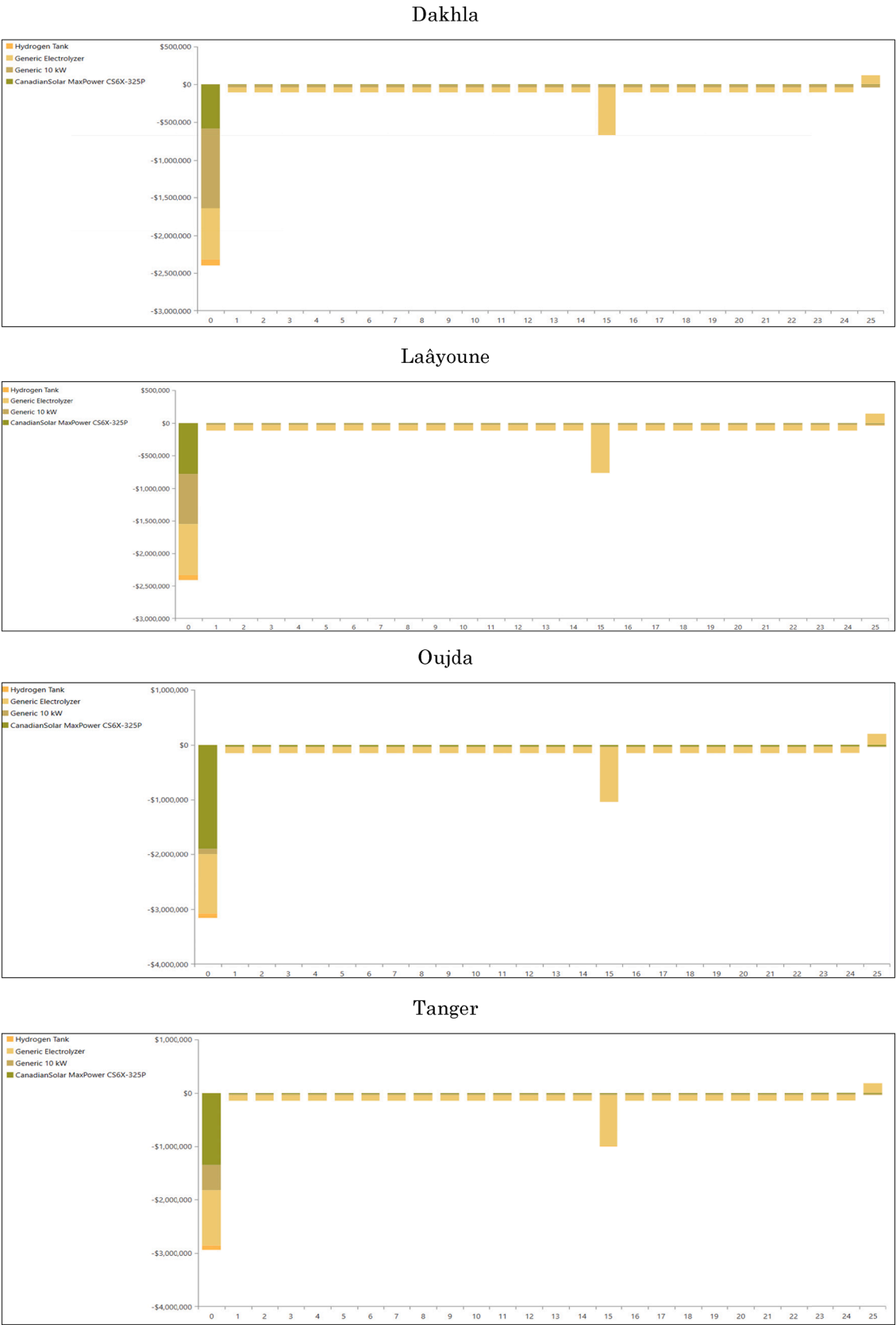


Fig. 12. Nominal cash flow of the system by component.

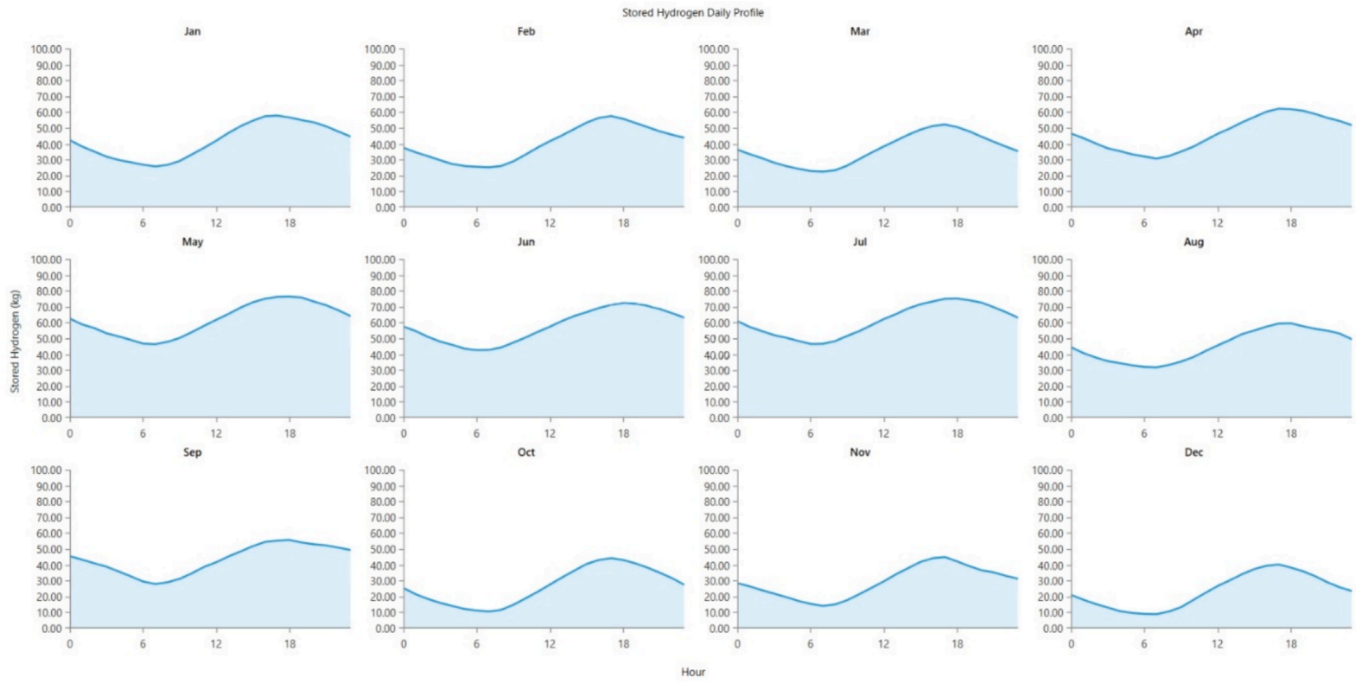


Fig. 13. The stored hydrogen daily profile over one year.

Table 8

LCOH for a 72 007 KG/h production unit of green hydrogen (Upscaling).

Regions	LCOH (\$/Kg)
Tanger-Tétouan-Al Hoceima (Tanger)	5.59
Oriental (Oujda)	6.95
Laâyoune-Sakia El Hamra (Laâyoune)	4.13
Dakhla-Oued Ed-Dahab (Dakhla)	3.89

wind resources, the optimization results indicate that PV is more economically attractive under the assumed cost and system parameters. This demonstrates that the optimal design of hybrid renewable systems depends not only on resource potential but also on economic, technical, and load related factors.

3.1.2. Analysis of economic indicators: NPC and LCOH

The Levelized Cost of Hydrogen 'LCOH' is a measure of the average

cost to produce hydrogen over the lifetime of a production facility, expressed in dollars per kilogram (\$/Kg). It is used to compare the economic feasibility of hydrogen production across different locations. Table 6 presents the results of LCOH for the four regions. Based on the results, Dakhla exhibits the lowest LCOH at 5.42\$/Kg, followed closely by Laâyoune at 5.57 \$/Kg. Tanger shows a higher LCOH of 6.86 \$/Kg, while the Oujda has the highest cost at 7.24 \$/Kg, indicating that southern regions have more favorable conditions for cost-efficient hydrogen production facilities compared to northern regions.

The obtained LCOH values, particularly for Dakhla-Oued Ed-Dahab (5.42 \$/kg), may appear higher when compared to certain Moroccan and international studies cited in the Introduction. However, it is important to emphasize that the present study considers a relatively small-scale hydrogen production capacity (7.28 kg/h), which limits the benefits of economies of scale. Electrolyzer capital cost per unit output is significantly higher at small capacities, leading to increased LCOH.

It is important to note that the reported LCOH values correspond to

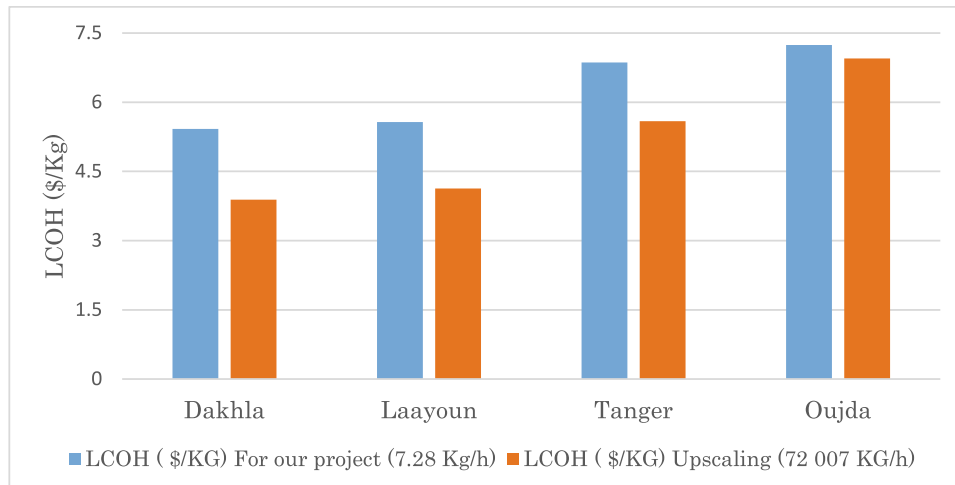


Fig. 14. Comparison of the LCOH for the SUSTEPS project and the upscaling scenario.

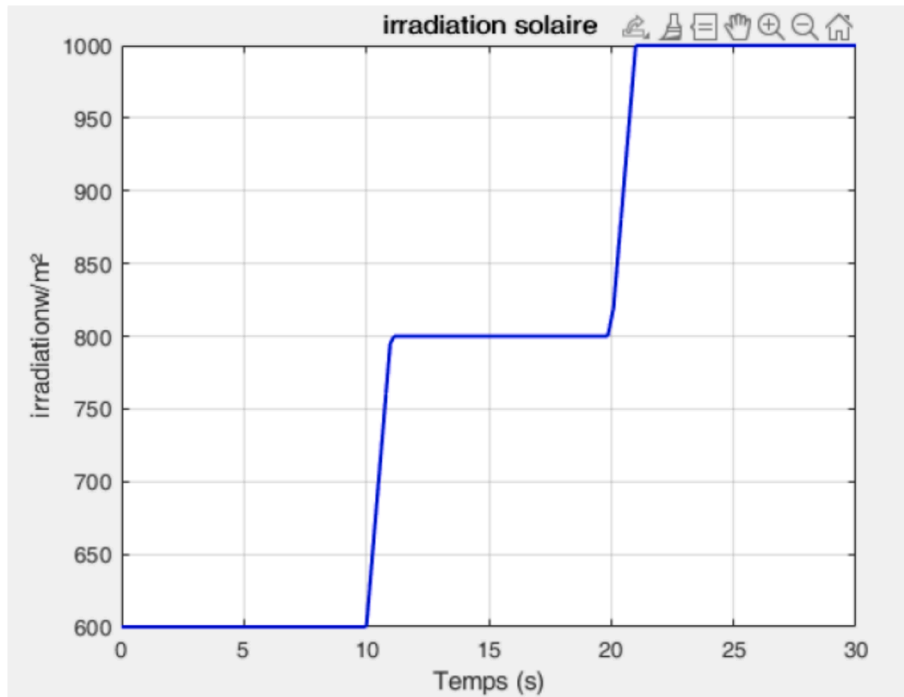


Fig. 15. Solar irradiance profile applied to the photovoltaic system during simulation.

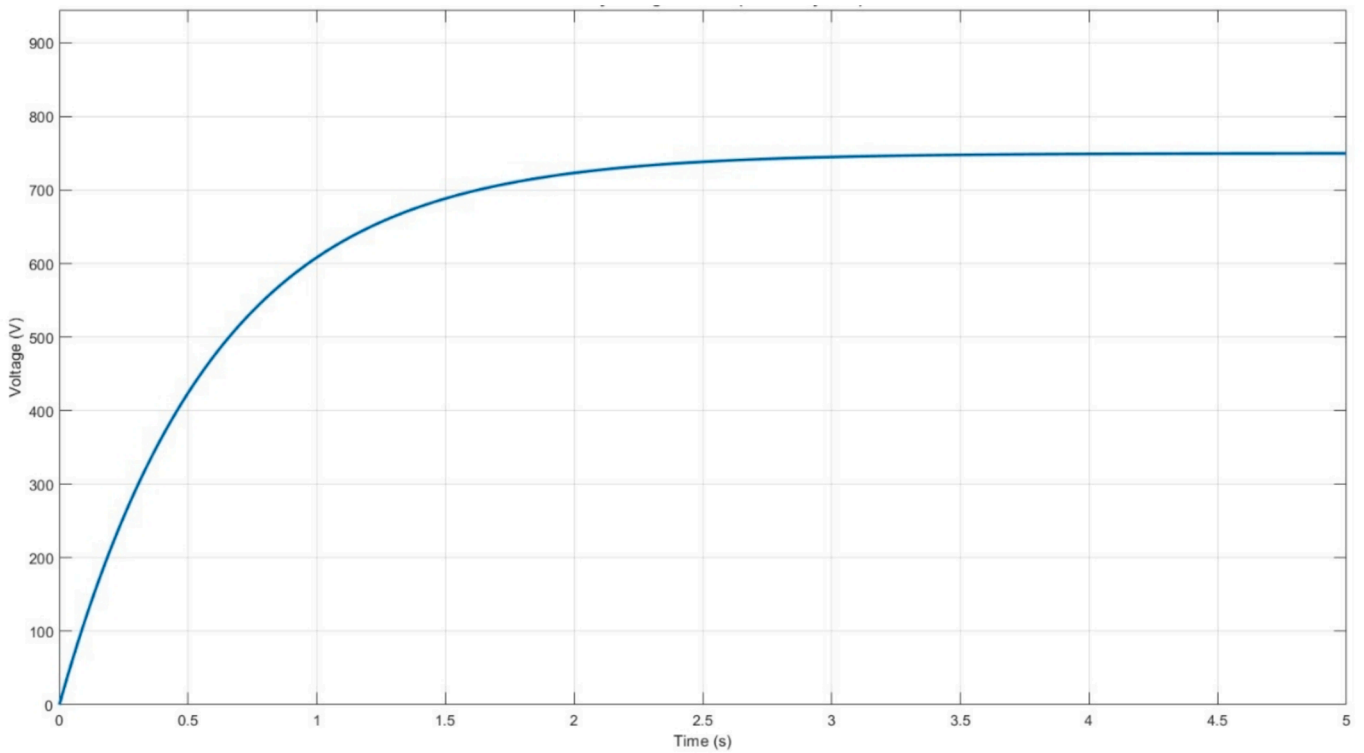


Fig. 16. Dynamic response of the photovoltaic system output voltage under varying solar irradiance conditions.

the core hydrogen production system and do not include auxiliary consumption. Incorporating these additional loads in future studies would provide a more comprehensive evaluation of system performance and economic cost.

Net Present Cost 'NPC' is a key economic indicator that reflects the total cost of a project over its entire lifetime, expressed in present value terms. It provides a useful basis for comparing economic feasibility

across different regions by accounting for both initial and ongoing costs. The results in Table 7 show that the Oriental region (Oujda) has the highest NPC at 5.37 million \$, followed by Tanger-Tétouan-Al Hoceima (Tanger) with 5.02 million \$. In contrast, Laâyoune-Sakia El Hamra (Laâyoune) and Dakhla-Oued Ed-Dahab (Dakhla) present lower NPCs of 4.13 million \$ and 4.02 million \$, respectively, indicating relatively lower overall investment requirements in these areas.

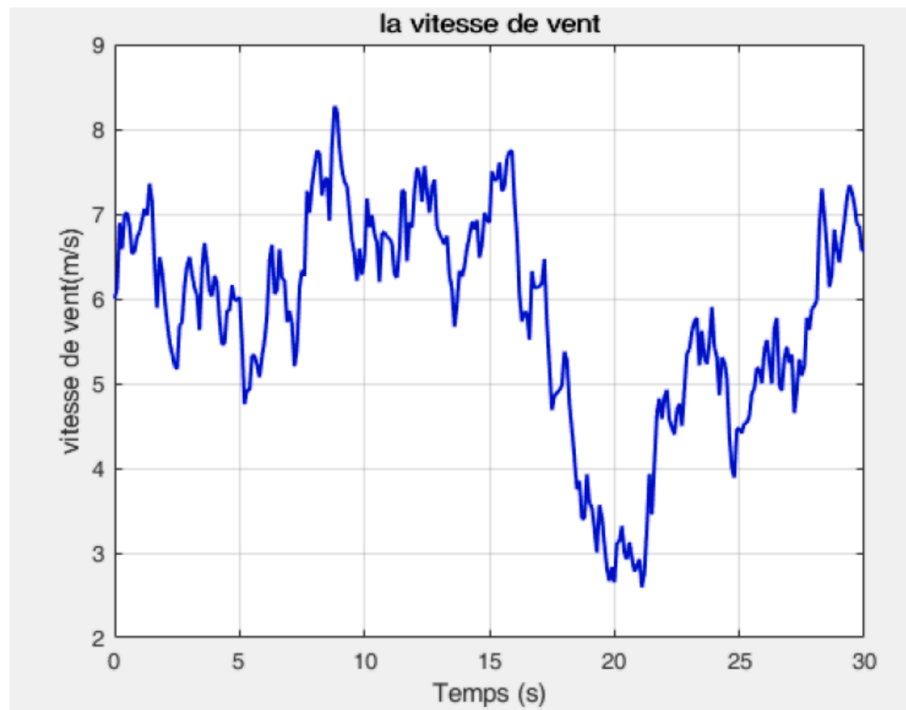


Fig. 17. Wind speed profile used for wind turbine simulation.

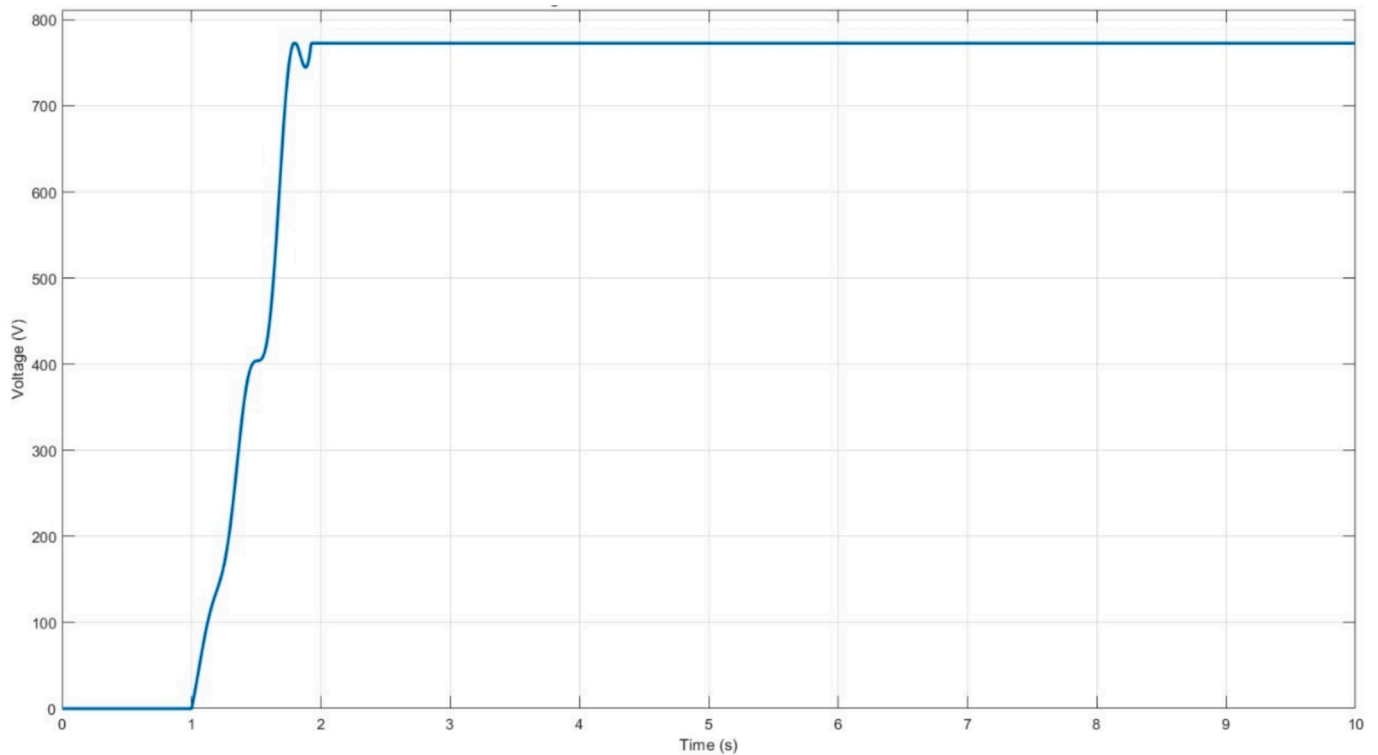


Fig. 18. Output voltage response of the wind energy system.

Fig. 12 shows the nominal cash flow of the system by component for the four regions.

In conclusion, based on the techno-economic results, Dakhla–Oued Ed-Dahab (Dakhla) appears to be the most suitable location for the proposed project.

3.1.3. Hydrogen storage

The HOMER Pro optimization identifies an optimal hydrogen storage capacity of 100 kg for all regions. Although battery storage is evaluated in the system design, the optimal configurations exclude batteries, favoring hydrogen storage.

This result can be explained by the lower cost of hydrogen storage

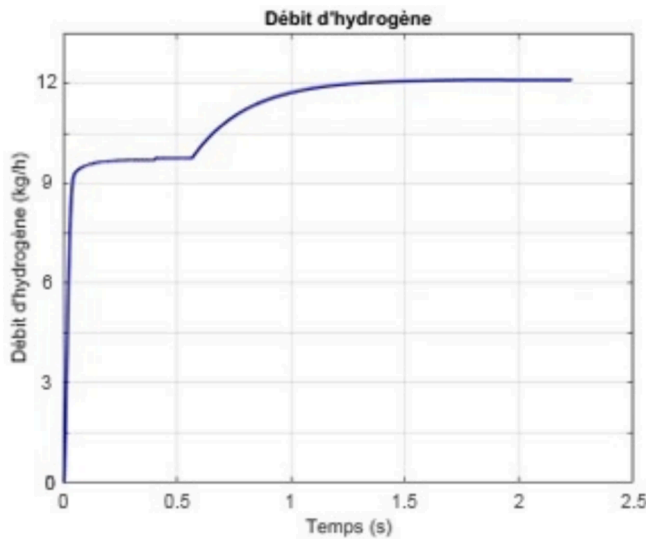


Fig. 19. Hydrogen production flow rate.

and the significantly higher NPC of batteries. Fig. 13 presents the stored hydrogen daily profile over one year. Storage levels are higher in May, June, and July, when solar and wind resources are at their peak. Given hydrogen's high flammability and low ignition energy, appropriate safety measures are essential for practical implementation. Although detailed risk assessment is beyond the scope of this study, following established safety standards is critical for the secure deployment of large-scale hydrogen systems.

3.1.4. Effect of project size on the LCOH

The results show that the Levelized Cost of Hydrogen (LCOH) in the Dakhla is lower than in Oujda, Laâyoune, and Tanger. However, it remains higher than the current global hydrogen production costs reported for regions with similar resource conditions. This can be attributed to the relatively small scale of the project, which has a hydrogen demand of 7.28 kg/h, while the capital cost of system components remains high.

Increasing the project scale, for instance to 72,007 tonnes per hour, would significantly reduce the LCOH. Therefore, the high LCOH observed in this study is justifiable given the limited system size. These results confirm that scaling up green hydrogen production leads to lower LCOH values, as larger electrolyzers benefit from economies of scale by distributing capital and operational costs over higher hydrogen output. This effect is particularly pronounced in regions with high renewable resources, such as Dakhla, making large-scale hydrogen production more economically competitive.

Table 8 illustrates the LCOH for 72,007 kg/h of hydrogen production, and Fig. 14 presents a comparison of the LCOH for the SUSTEPS project and the upscaling scenario.

3.2. Results and discussions from MATLAB Simulink

Fig. 15 presents the applied solar irradiance profile used to test the photovoltaic (PV) system, while Fig. 16 shows the corresponding PV system output voltage response.

The irradiance is varied in a stepwise manner, increasing from 600 W/m² to 800 W/m² at $t = 10$ s, and then to 1000 W/m² at $t = 20$ s. This profile is designed to emulate changing solar conditions and to evaluate the dynamic response and robustness of the PV model.

As illustrated in Fig. 16, the PV system voltage shows a rapid transient response at startup, followed by a stable operating regime. When the irradiance level increases, the PV voltage slightly rises and quickly reaches a new steady state value without significant oscillations.

These results confirm the correct implementation and consistency of the PV system model, demonstrating its ability to maintain stable electrical output under fluctuating irradiance conditions.

Fig. 17 illustrates the temporal variation of wind speed used as input for the wind turbine model. The wind profile is characterized by an average wind speed with random fluctuations added, representing realistic wind conditions and short-term variability. The wind speed varies approximately between 3 m/s and 8.5 m/s, allowing the evaluation of the dynamic response of the wind energy conversion system under changing operating conditions.

Fig. 18 presents the output voltage response of the wind energy system following power conversion and control. An initial transient is observed at the start of the simulation.

The voltage rapidly increases and stabilizes around 770 V, indicating the effective operation of the rectifier and DC/DC boost converter. Despite the fluctuations in wind speed, the output voltage remains stable after the transient phase, demonstrating the robustness of the control strategy and the effectiveness of the voltage regulation stage. The obtained results confirm that the proposed wind energy model is capable of capturing wind speed variability while ensuring a stable and regulated DC output.

Fig. 19 presents the hydrogen production flow rate as a function of time. Following a short transient phase, the hydrogen flow rate stabilizes at approximately 12 kg/h. Although the nominal hydrogen demand is 7.28 kg/h, the simulated production stabilizes around 12 kg/h due to system oversizing recommended by the HOMER Pro optimization. This strategy ensures reliability under renewable intermittency by allowing surplus hydrogen generation during high irradiance and wind periods. The excess hydrogen is stored in compressed gaseous form and later used to maintain the required continuous supply. Therefore, the higher production rate reflects a design choice for operational robustness rather than a deviation from the target demand.

4. Conclusion

In this paper, we presented a comprehensive techno-economic assessment and modeling of Green Hydrogen Production Using Renewable Energy resources across four Moroccan regions. The study began with a HOMER Pro-based optimization to identify the most suitable location for hydrogen production, where the Dakhla-Oued Ed-Dahab region emerged as the optimal site due to its superior renewable resource availability, achieving the lowest NPC and LCOH among the considered regions. Although the obtained LCOH remains higher than current global benchmarks, this was primarily attributed to the relatively small project scale and high capital costs. A scaling analysis demonstrated that increasing hydrogen production capacity leads to a significant reduction in LCOH, highlighting the importance of large-scale deployment for cost competitiveness. The system was then validated through detailed MATLAB/Simulink modeling of the PV system, wind energy conversion system, and alkaline electrolyzer, confirming stable electrical behavior under fluctuating solar irradiance and wind speed conditions. The electrolyzer achieved a steady hydrogen production rate of approximately 12 kg/h, exceeding the target demand of 7.28 kg/h, which supports using hydrogen storage as an effective energy reserve instead of battery storage.

Overall, the results demonstrate the technical feasibility, operational reliability, and economic relevance of hydrogen production from renewable sources in southern Morocco, reinforcing its potential role in supporting sustainable energy systems and future hydrogen deployment strategies.

The study is based on numerical simulations and techno-economic optimization using established tools and region-specific meteorological data to realistically represent system performance, while future work should focus on pilot scale experimental implementation to validate the system under real operating conditions. In addition, a detailed life cycle environmental assessment would provide a comprehensive

evaluation of the system's sustainability.

Furthermore, advancing green hydrogen requires interdisciplinary collaboration to ensure feasible, viable, and sustainable deployment.

CRedit authorship contribution statement

Omrane Derhy: Software, Methodology, Investigation, Data curation. **Tasme Bouanou:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis. **Nouhayla Atmani:** Validation, Project administration. **Loubna Bouslamti:** Software, Resources. **Meryeme Azaroual:** Resources, Methodology. **Nouhaila Nabil:** Visualization, Validation. **Ayoub Hirt:** Validation, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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